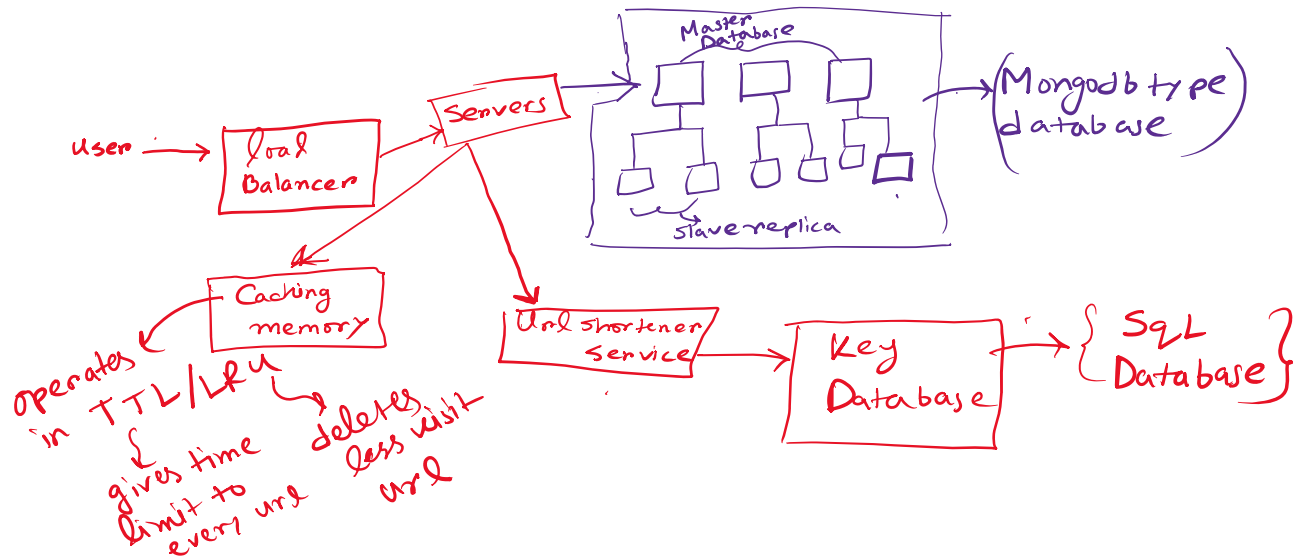


URL Shortener System

Sunday, May 31, 2026 9:07 PM



⊛ Here at first user inputs long url and rest api takes that long url as post command and Url shortening service retrieves a code against it from key data base. Then, long url, code & metadata is saved in database in sharding process and code made with url sent to user. Now again when user clicks it, api takes request as get, at first checks in cache memory, if no then goes to slave replica to get it.

Database: Here, master database store data & it replicates data to slave replica.

Now, for write master databases used & for read slave replicas used

This ensures long availability

Key Database:

Here all the codes are pre-generated.

"abc1234": True
"abc1567": False

Here codes pregenerated stored as keys in boolean. False means that code has been used or retrieved & true means code not been used yet. When user inputs URL shortening service pulls a code from this database.